

Lesson 13: Project - Hackathon

225 minutes

Overview

This is a five-day unit project. On day 1, students begin the project by choosing a partner, determining a dataset to design the app around, and creating a paper prototype. On Day 2, students continue to plan for the project by filling out tables of information on element IDs and programming constructs before each tackling a different role in the project as a designer or a programmer. Students spend days 3 and 4 continuing to build their apps. On the final day, students complete a Written Response, individually answering prompts about the project. Students then share their apps during a gallery walk.

Standards

Full Course Alignment

CSP Conceptual Framework

- **CRD-2** - Developers create and innovate using an iterative design process that is user-focused, that incorporates implementation/feedback cycles, and that leaves ample room for experimentation and risk-taking.

CSTA K-12 Computer Science Standards (2017)

- **AP** - Algorithms & Programming

Agenda

Day 1 (45 minutes)

Warm Up

Activity

Hackathon Project

Wrap Up

Assessment: Optional

Day 2 (45 minutes)

Warm Up

Activity

Wrap Up

Day 3 (45 minutes)

Warm Up

Activity

Wrap Up

Day 4 (45 minutes)

Objectives

Students will be able to:

- Complete a Written Response modeled after the Create PT
- Determine a dataset for project usage
- Effectively plan a project using a paper prototype
- Program an app which uses a database
- Translate a paper prototype to screens

Preparation

- Read through the Planning Guide, paying particular focus to the Scoring Guidelines
- Review the Example Student Projects, Planning Guides, and Written Responses under the "Resources" section
- Check the "**Teacher's Lounge**" forum for verified teachers to find additional strategies or resources shared by fellow teachers
- If you are teaching virtually, consider checking out the Virtual Modifications for Day 1, Day 2, Day 3, Day 4, and Day 5.

Links

Heads Up! Please make a copy of any documents you plan to share with students.

For the teachers

- **CSP Hackathon Submission on AP Classroom** - Resource

▼ Make a Copy

Warm Up

Activity

Wrap Up

Day 5 (45 minutes)

Warm Up

Activity

Wrap Up

- **Project - Hackathon** - Slides

▼ Make a Copy

- **Writing Support - Unit 6 Project - Hackathon** - Handout

▼ Make a Copy

For the students

- **CSP Hackathon Project Planning**

Guide - Activity Guide

▼ Make a Copy

- **CSP Hackathon Project Written**

Response - Handout ▼ Make a Copy

Teaching Guide

Day 1 (45 minutes)

Warm Up (3 minutes)

Remarks

Today we are beginning the Hackathon Project, which is an opportunity for you to show off what you've learned so far.

What is a hackathon? Traditionally, hackathons are events where people come together to build something creative in a short amount of time. Sometimes hackathons are centered around a goal or a problem to solve. Usually team members take on different roles (programming, designing, etc.) to get the project done within the time limit.

You will work with a partner on the Hackathon Project, using a database to create an interesting app.

Teaching Tip

Students will demonstrate their app design and programming skills throughout this five day project. In addition, students work with a dataset as this ensures students will be using the types of programming constructs required for the Create Performance Task, which this project is designed as a practice for. Students complete the project by individually filling out a Written Response, modeled after the Create PT.

This project can be used as a unit project, or as an end cap to the first semester of the course.

Activity (40 minutes)

Hackathon Prep

Teaching Tip

Forming Groups: You may opt to form the groups yourself, randomly place students in groups, or let students select their partners based on the dataset they want to work with.

Previewing the Written Responses: You may opt to share the [r cspu5written_response/csp/2021] with students early just so they know what they'll be expected to write at the end of the 5-day project.

 **Group:** Place students in groups of two.

Distribute: Pass out the **CSP U5 Hackathon Project Planning Guide** - one per group.

 **Read:** As a class, read through the Project Description on the Planning Guide.

Hackathon Project

 **Step 1:** Students navigate to a previous project and look at the dataset options. They choose a dataset that looks interesting to them.

 **Step 2:** Now students select how they will traverse a list pulled from their chosen dataset. Using the Planning Guide, students select whether they will use the filter, map, or reduce pattern and explain the specifics.

- Filter (most common option): use the list from one column to determine information that will be filtered from a list created by another column
 - Example: dogHeight filters dogNames, so only the names of small dogs are added to the filtered list
- Map: Add or change each item in a list
 - Example: map a list of numbers pulled from a column using Math.round - now each number is rounded
- Reduce: Reduce the data in a list to a single number
 - Example: find the smallest number in a list

 **Step 3:** The majority of the lesson should be spent on creating a detailed paper prototype. Students work together to draw out the screens of their app and design the flow from one screen to the next.

Wrap Up (2 minutes)

Remarks

In the next lesson, you will decide your project roles and we will begin to translate your paper prototype to the screen.

Assessment: Optional

Planning Guide: As an optional completion assessment, you can look at steps 1-3 in the Planning Guide.

Day 2 (45 minutes)

Warm Up

Do This: Move quickly to the activity portion of the lesson.

Activity (40 minutes)

Hackathon Project

 **Step 4:** Students work with their partners to fill out the tables on pages 4-6 in the Planning Guides. In these tables, students plan out all of the element IDs, onEvents, functions, variables, conditionals, and loops. This step, will help students smoothly transition into the programming phase.

 **Step 5 (3 mins):** On one computer, partners quickly build their screens, to be used by the programmer to test their code. Emphasize that these are NOT the final screens. Students drag out elements to the screen and give them the agreed upon names from Step 4. The design does not matter here.

 **Do This:** Partners select their primary roles.

- Programmer: Responsible for the majority of the programming. Needs to communicate decisions with the designer.
- Designer: Responsible for the design of the app. Pair programs with the programmer as needed.

 **Step 6:** Students work on building their apps, guided by their roles.

- Programmers: Use the table to guide you in adding programming statements to your project.
- Designers: Use the chart to guide you in adding screen elements to your program. You can work on a separate computer from your partner.

 Teaching Tip

If the designer finishes their screens early, they can start pair programming with the programmer.

Do This: When the designer is ready to share their screens with the programmer, there are specific steps that must be followed after the designer clicks share and copies the link to their app. The programmer:

1. Add a blank screen.
2. Delete the old project screens.
3. Click the screen dropdown, then click "Import screen"
4. Paste in the link from the designer.
5. Select to import all of the screens.
6. Delete the blank screen.
7. Set the home screen to be the default screen (Hint: Go to design mode and click on the screen)



Hackathon Project

Wrap Up (5 minutes)

Do This: Take some time to answer student questions.

Assessment: Optional

Planning Guide: As an optional completion assessment, you can look at the table in Step 4 of the Planning Guide.

Day 3 (45 minutes)

Warm Up

Do This: Move quickly to the activity portion of the lesson.

Activity (40 minutes)

Hackathon Project

Teaching Tip

Encourage students to check the rubric on the last page of the Planning Guide as they work through building their app. The Written Response portion will be completed on the last day of the project.

Step 6 (continued): Students work on building their apps, guided by their roles.

- Programmers: Use the table to guide you in adding programming statements to your project.
- Designers: Use the chart to guide you in adding screen elements to your program. You can work on a separate computer from your partner.

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If the designer finishes their screens early, they can start pair programming with the programmer.

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Wrap Up (5 minutes)

Do This: Take some time to answer student questions.

Assessment: Optional

Planning Guide: As an optional completion assessment, you can look at the table in Step 4 of the Planning Guide.

Day 4 (45 minutes)

Warm Up

Do This: Move quickly to the activity portion of the lesson.

Activity (40 minutes)

Hackathon Project

Step 6 (continued): Students work on building their apps, guided by their roles.

- Programmers: Use the table to guide you in adding programming statements to your project.

- Designers: Use the chart to guide you in adding screen elements to your program. You can work on a separate computer from your partner.

💡 Teaching Tip ▲

If the designer finishes their screens early, they can start pair programming with the programmer.

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6. Delete the blank screen.
7. Set the home screen to be the default screen (Hint: Go to design mode and click on the screen)

📋 Do This: Take some time around the midpoint of the activity for students to check the Overall Project section of the Scoring Guidelines (final page of the Planning Guide).

Wrap Up (5 minutes)

📋 Do This: Remind students that they will be completing the Written Response portion of the project *individually* during the next lesson. Both students need to fully understand how the program works and have access to the final project.

Assessment Opportunity: Optional

Planning Guide: As an optional completion assessment, you can look at the table in Step 4 of the Planning Guide.

Day 5 (45 minutes)

Warm Up

Do This: Move quickly to the activity portion of the lesson.

Activity (40 minutes)

Written Response

Note: The Written Response is to be completed individually. From this point on, partners should not discuss anything about their project.

💡 Teaching Tip ▲

The Written Response portion of this project is a modified, short version of the Create Performance Task students will turn in to the College Board for the AP Exam. To have your students practice submitting this project using a similar interface to the Create Performance Task, see the [**CSP Hackathon Submission on AP**](#)

Classroom for instructions on how to set up an assessment on AP Classroom.

 **Do This:** All students individually navigate to their projects on one tab or browser window. In another tab, students open the Written Response.

 Teaching Tip

Programmers should share the project link with the designers. Designers open the project, then click "View Code", which is located on the top right of the screen.

 Remarks

 On your Written Response, you will need to insert screenshots of code segments. What's a code segment? It's a collection of program statements that are part of a program. Together, they demonstrate some working part of your program. You can use screenshots of text or blocks for your code segments.

 Teaching Tip

Students may struggle with the prompt on managing complexity. Encourage students to think how the program would be written differently if no lists could be used. In most cases, this would involve a large amount of variables. Students need to be specific and explain what these variables would store.

 **Do This (30 mins):** Students work silently to complete the Written Response. Students will need to take screenshots of code and insert those screenshots into the appropriate boxes in the Written Response. Students can check the Scoring Guidelines in the Planning Guide to make sure they are meeting all the requirements.

 Teaching Tip

Each student, in every classroom, will come to the written response with a different level of comfort, ability, and familiarity with writing for these style of questions. As such, consider offering one or more of the following supports:

Writing Support: Provide students with these sentence starters for guiding their response.

App Code Support: For students who did not fully complete their app, provide them with this adaptation of the exemplar code.

 **Submit:** Students submit their project (one per group), Planning Guide (one per group), and Written Response (individual).

 **Share:** Now students display their projects on a computer and complete a gallery walk. If you decide to have a winner of the hackathon, students can vote on their favorite projects by writing down the name of the project on a sticky note and passing it in.



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Hackathon Project

Wrap Up (5 minutes)

 Teaching Tip

Feel free to skip the announcement of a hackathon winner if you want to keep the project non-competitive.

Remarks

Great work everyone on the Hackathon Project! It's amazing how much you were able to accomplish in a short amount of time. And now, the moment that everyone has been waiting for. I have tallied up the votes and the winner is...

Assessment: Grading the Project

Project: Use the Scoring Guidelines on the final page of the Planning Guide to assess student projects. The Written Response portion is individual, while the Overall Project portion is a group grade.



End of Unit Survey



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